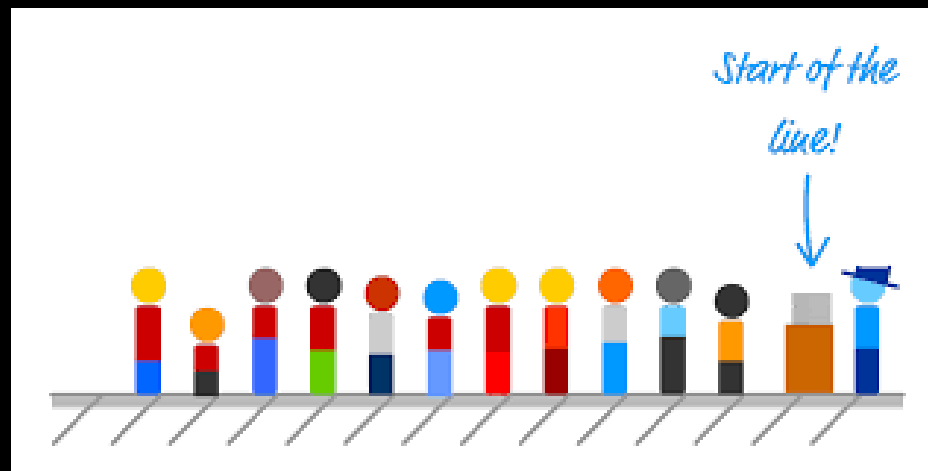


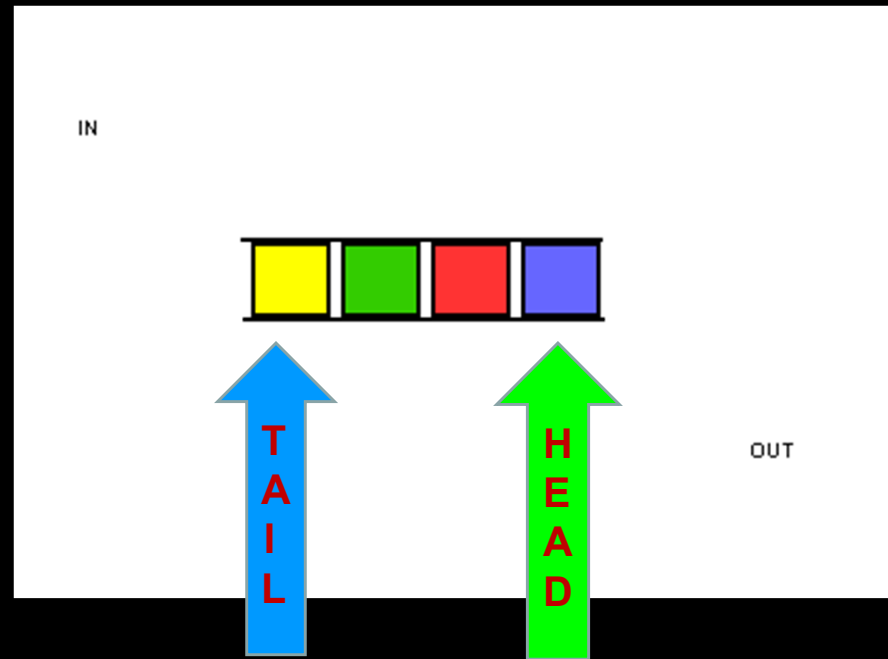
LT10c QUEUE

What is a **queue**?

- A queue is an **ordered** collection of items.
- It is a **first in first out (FIFO)** data structure.
- Elements are usually removed in the same order as which they are added.



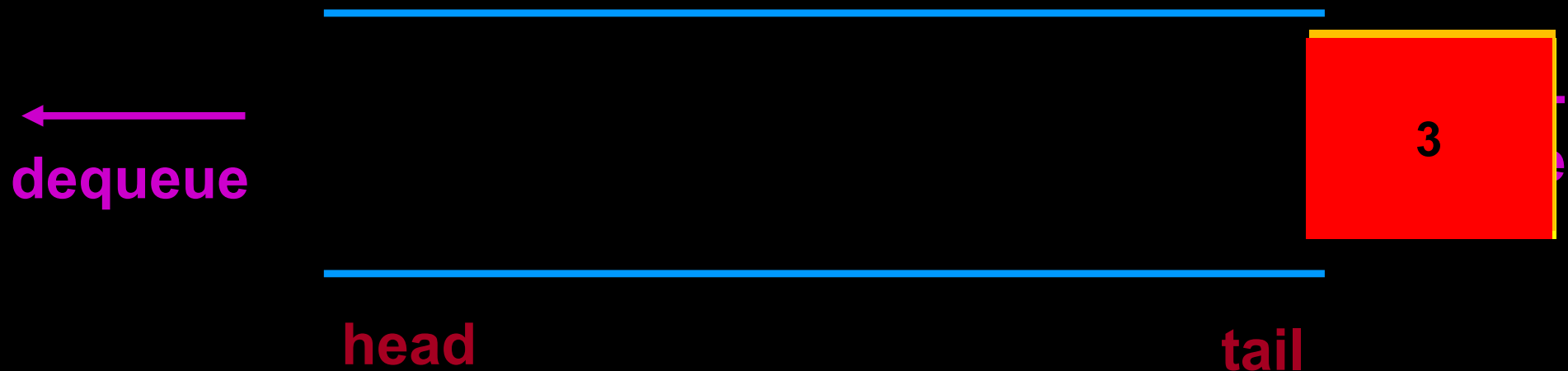
What is a **queue**?



An element is **added** to the **tail** position of the queue.

An element is **removed** from the **head** of the queue.

Implementation of a FIFO queue

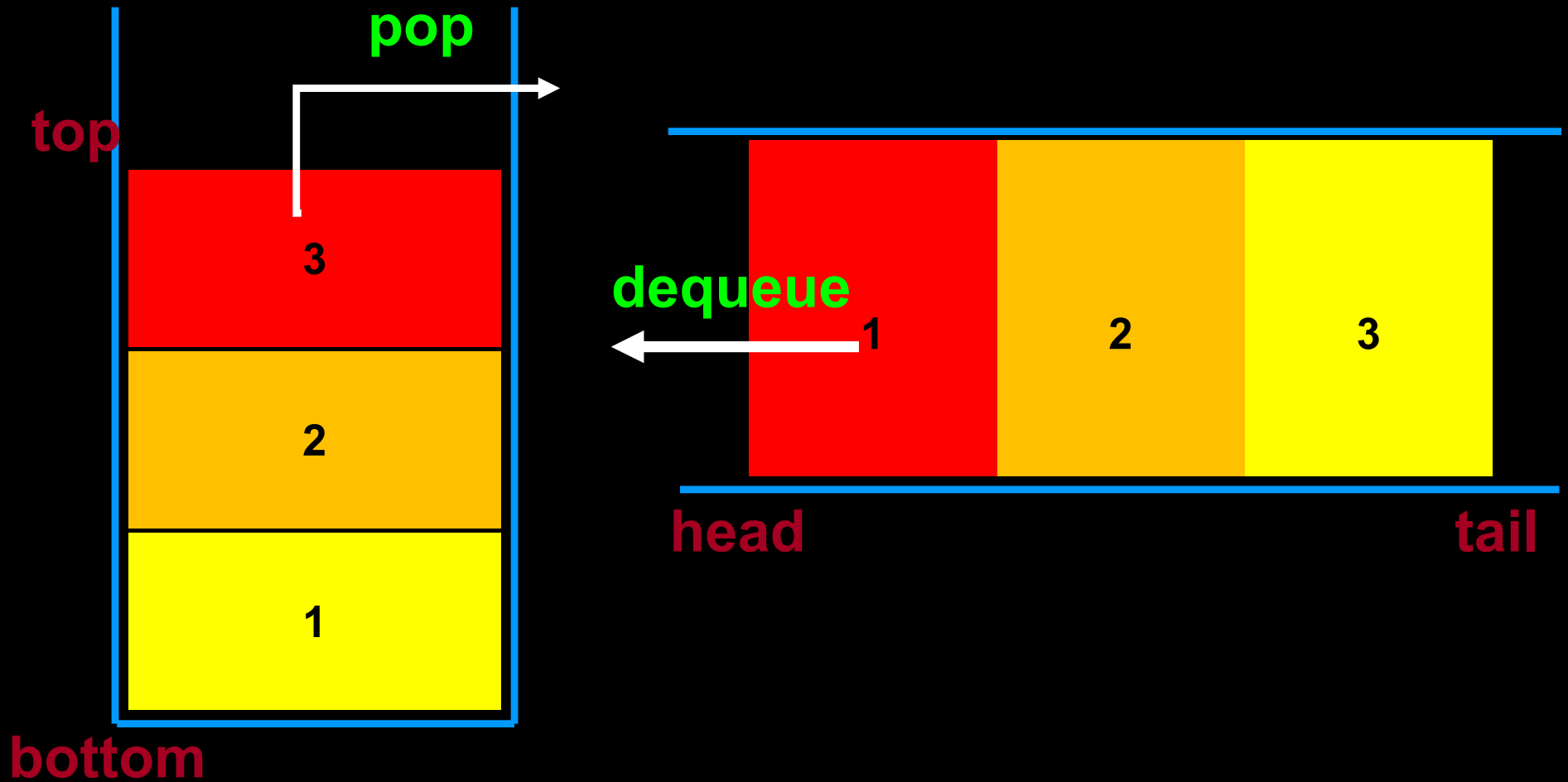


Main Functions

enqueue / push

dequeue / pop

What's the difference?



Stack vs Queue

You will be implementing the queue ADT in lecture training:

Queue ADT	Description
make_queue()	returns a new, empty queue
enqueue(queue, item)	adds item to queue
dequeue(queue)	removes the first item from queue and returns it
front(queue)	returns the first object of the queue but does not remove it from the queue
is_empty(queue)	returns True if queue is empty, False if otherwise

List Operations for Queue

Queue Operation	List Operation	Description
enqueue	<code>lst.append(item)</code>	Modifies list by adding an item to the end of the list
dequeue	<code>lst.pop(i)</code>	Removes the item at index i and returns it

Dequeue should remove the element from the front of the queue:
use `lst.pop(0)`.

Implement a FIFO queue using a list

```
q = make_queue()
```

```
dequeue(q)           → None : empty queue,  
enqueue(q, 5)        nothing to dequeue
```

```
enqueue(q, 3)
```

```
dequeue(q)           → 5
```

```
dequeue(q)           → 3
```

```
is_empty(q)          → True
```


Applications of Queue (Tutorial)

- ▶ Scheduling of processes
- ▶ Network printer queue
- ▶ Keyboard buffer

Summary

- A queue is an **ordered** collection of items.
- Used when the order needs to be preserved
- A queue is **FIFO** data structure
- Typical operations include enqueue (add items) and dequeue (remove items)

Stack vs Queue

STACK	QUEUE
Follows the LIFO order	Follows the FIFO order
Operations are push and pop	Operations are enqueue and dequeue
pop removes an item from the top of the stack	dequeue removes an item from the head of the queue
peek retrieves an item from the top of the stack	front retrieves an item from the head of the queue
A stack is a vertical collection of items	A queue is a horizontal collection of items